

COMP 8005

Assignment 3

Testing

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Test	Expected	Actual	Screenshot
No arguments	fail	fail	Test 1
-h	pass	pass	Test 2
-x	fail	fail	Test 3
Missing argument	fail	fail	Test 4
Negative port	fail	fail	Test 5
Too many arguments	fail	fail	Test 6
Shadow file doesn't exist	fail	fail	Test 7
User doesn't exist	fail	fail	Test 8
MD5 password (1 thread)	pass	pass	Test 9
bcrypt password (1 thread)	pass	pass	Test 10
SHA-256 password (1 thread)	pass	pass	Test 11
SHA-512 password (1 thread)	pass	pass	Test 12
yescrypt password (1 thread)	pass	pass	Test 13
Worker Invalid IP Address	fail	fail	Test 14
Worker heartbeats	pass	pass	Test 15
Controller no heartbeat response	fail	fail	Test 16
Multiple workers	pass	pass	Test 17

Tests

Test 1

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py
usage: controller.py [-h] -f FILE -u USER -p PORT -b HEARTBEAT_SECONDS
controller.py: error: the following arguments are required: -f/--file, -u/--user, -p/--port, -b/--heartbeat_seconds
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ s
```

Test 2

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -h
usage: controller.py [-h] -f FILE -u USER -p PORT -b HEARTBEAT_SECONDS -c CHUNK_SIZE

Password Cracker Controller Script

options:
  -h, --help            show this help message and exit
  -f FILE, --file FILE  Path to the shadow file
  -u USER, --user USER  Username of the password to crack
  -p PORT, --port PORT  Port number to host on
  -b HEARTBEAT_SECONDS, --heartbeat_seconds HEARTBEAT_SECONDS
                        Interval between heartbeat requests (seconds)
  -c CHUNK_SIZE, --chunk_size CHUNK_SIZE
                        Chunk size to send workers
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 3

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -x
usage: controller.py [-h] -f FILE -u USER -p PORT -b HEARTBEAT_SECONDS
controller.py: error: the following arguments are required: -f/--file, -u/--user, -p/--port, -b/--heartbeat_seconds
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 4

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -f ./shadow -u user1 -p 8080
usage: controller.py [-h] -f FILE -u USER -p PORT -b HEARTBEAT_SECONDS
controller.py: error: the following arguments are required: -b/--heartbeat_seconds
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 5

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -f ./shadow -u user1 -p -8080 -h 2
usage: controller.py [-h] -f FILE -u USER -p PORT -b HEARTBEAT_SECONDS
controller.py: error: argument -p/--port: -8080 is not a positive integer
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 6

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -f ./bad_file -u user1 -p 8080 -b 2
usage: controller.py [-h] -f FILE -u USER -p PORT -b HEARTBEAT_SECONDS
controller.py: error: unrecognized arguments: 1
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 7

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -f ./bad_file -u user1 -p 8080 -b 2
Error: Failed to open shadow file.
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 8

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -f ./shadow -u fake -p 8080 -b 2
Error: Username not found in shadow file.
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 9

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -f ./shadow -u user1 -p 8080 -b 2
Shadow file line:
user1:$1$VDu0qEN8$7dBMep08zXS.0QG4kkLE//:20474:0:99999:7:::
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Connection received: ('::ffff:127.0.0.1', 43462, 0, 0)
Sent data to ('::ffff:127.0.0.1', 43462, 0, 0)
Sent heartbeat request to ('::ffff:127.0.0.1', 43462, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 0}
[HEARTBEAT] Response received, 0 attempts tried.
Sent heartbeat request to ('::ffff:127.0.0.1', 43462, 0, 0)
Received data from client: {'type': 'password', 'password': 'abc', 'crack_time': datetime.timedelta(microseconds=605676)
6, 2, 10, 15, 36, 15, 657805)}
=====
Hash Algorithm: MD5
Password Found: abc
Controller Parsing Time: 6e-05 seconds
Dispatch Latency: 0.000108 seconds
Cracking Time: 0.605676 seconds
Return Latency: 1.126527 seconds
Total end-to-end Runtime: 1.732371 seconds
=====
```

Test 10

```
[HEARTBEAT] Response received, 1157 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 57844, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 2492}
[HEARTBEAT] Response received, 2492 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 57844, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 3756}
[HEARTBEAT] Response received, 3756 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 57844, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 5054}
[HEARTBEAT] Response received, 5054 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 57844, 0, 0)
Received data from client: {'type': 'password', 'password': 'abc', 'crack_time': datetime.timedelta(seconds=9, microseconds=499030)}
atetime(2026, 2, 10, 15, 37, 11, 499030)}
=====
Hash Algorithm: bcrypt
Password Found: abc
Controller Parsing Time: 6.6e-05 seconds
Dispatch Latency: 0.000108 seconds
Cracking Time: 9.710764 seconds
Return Latency: 0.013404 seconds
Total end-to-end Runtime: 9.724341999999998 seconds
=====
```

Test 11

```
[HEARTBEAT] Response received, 0 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 36846, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 1404}
[HEARTBEAT] Response received, 1404 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 36846, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 3151}
[HEARTBEAT] Response received, 3151 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 36846, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 4867}
[HEARTBEAT] Response received, 4867 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 36846, 0, 0)
Received data from client: {'type': 'password', 'password': 'abc', 'crack_time': datetime.timedelta(seconds=7, microseconds=153518)}
atetime(2026, 2, 10, 15, 37, 40, 153518)}
=====
Hash Algorithm: SHA-256
Password Found: abc
Controller Parsing Time: 5.3e-05 seconds
Dispatch Latency: 0.000138 seconds
Cracking Time: 7.538771 seconds
Return Latency: 0.217841 seconds
Total end-to-end Runtime: 7.756803 seconds
=====
```

Test 12

```
[HEARTBEAT] Response received, 1139 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 38692, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 2426}
[HEARTBEAT] Response received, 2426 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 38692, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 3752}
[HEARTBEAT] Response received, 3752 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 38692, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 5110}
[HEARTBEAT] Response received, 5110 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 38692, 0, 0)
Received data from client: {'type': 'password', 'password': 'abc', 'crack_time': datetime.timedelta(seconds=9, microseconds=424291)}
tetime(2026, 2, 10, 15, 38, 26, 424291)}
=====
Hash Algorithm: SHA-512
Password Found: abc
Controller Parsing Time: 0.000171 seconds
Dispatch Latency: 9.9e-05 seconds
Cracking Time: 9.705055 seconds
Return Latency: 0.06463 seconds
Total end-to-end Runtime: 9.769955 seconds
=====
```

Test 13

```
Sent heartbeat request to (::ffff:127.0.0.1', 58026, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 6180}
[HEARTBEAT] Response received, 6180 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 58026, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 6301}
[HEARTBEAT] Response received, 6301 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 58026, 0, 0)
Received data from client: {'type': 'password', 'password': 'abc', 'crack_time': datetime.timedelta(seconds=95, microseconds=10423),
atime(2026, 2, 10, 15, 40, 20, 77916)}
=====
Hash Algorithm: yescrypt
Password Found: abc
Controller Parsing Time: 8.4e-05 seconds
Dispatch Latency: 0.000104 seconds
Cracking Time: 95.010423 seconds
Return Latency: 0.384088 seconds
Total end-to-end Runtime: 95.394699 seconds
=====
```

Test 14

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005$ python3 ./source/worker.py -c bad_ip -p 8080 -t 1
/home/wonton/Desktop/COMP8005/./source/worker.py:5: DeprecationWarning: 'crypt' is deprecated and slated for removal in Python 3.13
import crypt # Deprecated, but used for yescrypt
Error: Invalid IP Address found.
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005$
```

Test 15

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005$ python3 ./source/worker.py -c 127.0.0.1 -p 8080 -t 1
/home/wonton/Desktop/COMP8005/./source/worker.py:5: DeprecationWarning: 'crypt' is deprecated and slated for removal in Python 3.13
import crypt # Deprecated, but used for yescrypt
Received response
{'type': 'job', 'hash_algorithm': '6', 'salt': '$X90hJRFVxXSiqImk$', 'hashed_password': 'KKIN7ssYbwXz9WJgKxY7zNcg03ctu1snY5MjqoQgVpQY
e.datetime(2026, 2, 10, 15, 38, 16, 432743)}
Populating queue
[HEARTBEAT] Request received
[HEARTBEAT] Response sent
Finished populating queue
[HEARTBEAT] Request received
[HEARTBEAT] Response sent
[HEARTBEAT] Request received
[HEARTBEAT] Response sent
[HEARTBEAT] Request received
[HEARTBEAT] Response sent
[HEARTBEAT] Request received
[HEARTBEAT] Response sent
[HEARTBEAT] Request received
[HEARTBEAT] Response sent
[FOUND] Password: abc
Total attempts: 6402
```

Test 16

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$ python3 ./controller.py -f ./shadow -u user13 -p 8080 -b 2
Shadow file line:
user13:$y$j9T$HjgMvmSW0Ii.IpKGM9JHq1$MQXJaa0/zzkxdG1NzeCt3U4QeGPQEt0kiTwnlU4J.z2:20474:0:99999:7:::
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Connection received: (::ffff:127.0.0.1', 35138, 0, 0)
Sent data to (::ffff:127.0.0.1', 35138, 0, 0)
Sent heartbeat request to (::ffff:127.0.0.1', 35138, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 0}
[HEARTBEAT] Response received, 0 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 35138, 0, 0)
Received data from client: {'type': 'heartbeat', 'attempts': 144}
[HEARTBEAT] Response received, 144 attempts tried.
Sent heartbeat request to (::ffff:127.0.0.1', 35138, 0, 0)
[HEARTBEAT] Failed to receive heartbeat response, shutting down
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005/source$
```

Test 17

```
wonton@wonton-IdeaPad-Slim-7-14ITL05:~/Desktop/COMP8005$ python3 ./source/controller.py -p 8080 -f ./source/shadow -c 10000 -b 5 -u user6
Shadow file line:
user6:$2b$05$NUDWAxv5uIdv/GVEInudl.Q7EB3MLLYovfSmvx96Qb9vBKl0t/wu:20474:0:99999:7:::
Server running on dual-stack mode (IPv4 and IPv6).
Server is listening on port 8080 for incoming connections...
Worker registered: ('::ffff:192.168.1.73', 64246, 0, 0)
Worker registered: ('::ffff:192.168.1.99', 50296, 0, 0)
█
```

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